# UNIT 2

# Creative Thinking and Design Ideation

**Computational Design Thinking – Detailed B.Tech Notes**

**Topics:** Creative Thinking in Computational Design; Divergent & Convergent Thinking; Mind Mapping; Brainstorming & Brainwriting; SCAMPER; Idea Generation; Lateral Thinking Strategies; Design Thinking Process – Empathize, Define, Ideate, Selecting & Prioritizing Ideas.

## 1. Creative Thinking in Computational Design

**Creative thinking** is the ability to look at a problem in new, useful and different ways and generate ideas that can lead to better solutions. In computational design, creativity is combined with logical and systematic thinking to design solutions for real-world problems.

**Computational design** uses computational methods such as algorithms, data, logical reasoning, models and systematic processes to understand problems and create solutions. Creativity helps decide *what* solution could be useful, while computational thinking helps determine *how* it can be designed and implemented.

### Importance of creative thinking:

- Helps generate multiple possible solutions instead of stopping at the first solution.
- Encourages improvement of existing products, services and processes.
- Helps deal with unfamiliar and ill-structured problems.
- Supports innovation in software, products, services and workflows.
- Connects technical possibilities with real user needs.
- Can lead to simpler, faster, safer, more accessible or cost-effective solutions.

**Example:** If students cannot easily find empty classrooms, a conventional solution may be asking the office. A creative computational solution could be a system that displays classroom availability using timetable/database information and allows searching by building, floor and time.

**Key point:** Creative thinking is not simply random thinking. A strong creative idea should generally be both **novel** (different) and **useful** (relevant).

## 2. Divergent and Convergent Thinking

**Divergent thinking** expands the problem space by generating many different, varied and unconventional ideas without immediately judging them.

**Convergent thinking** narrows the generated ideas by evaluating them against requirements, constraints and criteria and selecting the most suitable option.

**Easy formula:** Divergent = **Generate many**; Convergent = **Select the best**.

**Typical sequence:** Problem → Divergent Thinking → Many Ideas → Convergent Thinking → Evaluation → Selected Idea → Prototype/Solution.

### Important points:

- During divergent thinking, postpone criticism and evaluation.
- Encourage variety, not only small changes to one idea.

- During convergence, use clear criteria instead of personal preference alone.
- Two or more ideas can be combined before final selection.

| Divergent Thinking | Convergent Thinking |
|---|---|
| Expands possibilities | Narrows possibilities |
| Focus: quantity and variety | Focus: quality and suitability |
| Judgement is postponed | Judgement is required |
| Question: What could we do? | Question: Which should we choose? |

# 3. Mind Mapping for Idea Generation

A **mind map** is a visual technique for organizing information and generating related ideas around a central topic. The main problem or concept is placed at the centre and branches represent related categories, sub-problems, features or possible solutions.

## Steps:

- Write the central problem or topic in the middle.
- Create major branches such as users, causes, requirements, features, risks or solutions.
- Add smaller branches containing related ideas.
- Use short keywords where possible.
- Look for connections between different branches.
- Use the map to discover opportunities and generate solution ideas.

**Benefits:** improves visual organization, encourages associations, reveals relationships, supports memory and helps teams explore a problem broadly.

**Example:** Central topic: “Improve campus navigation.” Branches could include Maps, Accessibility, Buildings, Notifications, Transport, Search and Emergency Help. Each branch can produce further ideas.

# 4. Brainstorming

**Brainstorming** is a group idea-generation technique in which participants produce many ideas around a defined problem before detailed evaluation begins.

## Main rules:

- Defer judgement: do not criticize ideas while ideas are being generated.
- Focus on quantity: many ideas increase the chance of finding a strong solution.
- Encourage unusual ideas: unusual ideas can lead to useful combinations.
- Build on others’ ideas: improve, combine or extend suggestions.
- Keep the problem statement clear so ideas remain relevant.

**Process:** Define challenge → Explain rules → Generate ideas → Record all ideas → Group similar ideas → Evaluate later.

**Common mistakes:** judging too early, allowing one person to dominate, moving away from the problem, repeating ideas, or selecting the first acceptable idea.

# 5. Brainwriting

**Brainwriting** is an idea-generation method in which participants write their ideas instead of speaking them aloud. It can reduce the effect of dominant speakers and give quieter participants an equal opportunity to contribute.

**Typical process:** Present problem → participants independently write ideas → ideas are shared → participants build on ideas → group evaluates them.

**Brainstorming vs Brainwriting:** Brainstorming is primarily verbal and interactive; brainwriting is primarily written and can give more equal participation.

**Useful when:** the group is large, participants are shy, the activity is online, time is limited, or independent thinking is preferred.

# 6. SCAMPER Technique

**SCAMPER** is a structured creativity technique used to modify or improve an existing product, service, process or idea. It provides seven prompts for examining an existing solution from different perspectives.

## S – Substitute

Replace a component, material, person, step, method or feature. Ask: “What can be replaced?”

## C – Combine

Merge two or more features, functions, steps or ideas. Ask: “What can be combined?”

## A – Adapt

Borrow or adjust an existing idea so it works in a new situation. Ask: “What can be adapted?”

## M – Modify / Magnify / Minify

Change size, shape, appearance, speed, intensity or another property. Ask: “What can be changed, enlarged or reduced?”

## P – Put to Another Use

Use an existing product, resource or idea for a different purpose. Ask: “Can it be used somewhere else?”

## E – Eliminate

Remove unnecessary steps, features, components or complexity. Ask: “What can be removed?”

## R – Reverse / Rearrange

Reverse the order, direction or role of elements, or rearrange them. Ask: “What happens if we change the sequence?”

**Example – Student learning portal:** Substitute a complex login method; Combine attendance and assignment notifications; Adapt a familiar mobile-notification model; Modify the dashboard layout; Put the notification system to another use for emergency announcements; Eliminate unnecessary clicks; Rearrange navigation according to frequently used functions.

# 7. Idea Generation

**Idea generation** is the process of producing possible solutions, approaches or concepts in response to a problem or opportunity. At the beginning, the objective is not to find one perfect answer but to explore a useful range of possibilities.

## Sources of ideas:

- User observations and interviews
- Existing products and their limitations
- Complaints, pain points and repeated difficulties
- Research, data and trends
- Brainstorming and brainwriting
- Mind maps and SCAMPER
- Analogies and solutions from other domains
- Technology capabilities and emerging tools

**Good practice:** understand the problem first, generate alternatives, record ideas, combine related ideas and postpone detailed judgement until sufficient alternatives have been produced.

# 8. Lateral Thinking Strategies

**Lateral thinking** is an approach to problem solving that deliberately moves away from conventional or habitual patterns and looks for alternative perspectives, assumptions and routes to a solution.

## Strategies:

- **Challenge assumptions:** identify what people take for granted and ask whether it must remain true.
- **Reverse the approach:** consider doing the opposite of the usual process.
- **Change perspective:** examine the problem from another user's, stakeholder's or system's viewpoint.
- **Break fixed patterns:** question the normal sequence, arrangement or workflow.
- **Random stimulation:** introduce an unrelated word/object/concept and search for useful connections.
- **Provocation:** deliberately propose an unusual statement to trigger a new direction of thought.
- **Reframing:** rewrite the problem differently so that new solution directions become visible.

**Example:** If a website is slow, the conventional approach may be "increase server capacity." Lateral thinking asks: Can we reduce unnecessary data? Can we change loading order? Can content load only when requested? Can the user task be redesigned so less information is needed at once?

# 9. Design Thinking Process

Design thinking is a human-centered approach to solving problems. For this unit, the main stages are **Empathize** → **Define** → **Ideate** → **Select & Prioritize Ideas**. The process is iterative, so teams can return to an earlier stage when new information reveals a problem.

## Stage 1 – Empathize

Empathize means understanding users, their needs, experiences, difficulties, expectations and context rather than assuming what they need.

- Observe users and their behaviour.
- Conduct interviews or surveys.
- Listen for pain points and unmet needs.
- Understand emotions, goals and constraints.
- Avoid deciding the solution before understanding the user.

**Output:** user insights, observations, pain points and a clearer understanding of the user.

### Stage 2 – Define

Define means converting research and observations into a clear, focused problem statement. A good problem statement identifies the user, need and context without unnecessarily prescribing the solution.

**Example:** Instead of “We need to create a campus app,” define the need as: “Students need a simple way to find available classrooms because they currently depend on scattered information.”

### Stage 3 – Ideate

Ideate means generating a broad range of possible solutions. Brainstorming, brainwriting, mind mapping, SCAMPER and lateral thinking can be used here.

Use divergent thinking first. Generate alternatives before using convergent thinking to evaluate them.

### Stage 4 – Selecting and Prioritizing Ideas

Selection means evaluating alternatives against agreed criteria. Prioritization means deciding which ideas deserve attention first based on factors such as user value, feasibility, cost, time, risk and technical requirements.

## 10. Selecting and Prioritizing Ideas

**Do not select an idea only because it sounds interesting.** Use clear evaluation criteria:

- **User value:** Does it solve an important user problem?
- **Feasibility:** Can it realistically be implemented with available technology and skills?
- **Cost:** Is the budget reasonable?
- **Time:** Can it be developed within the available time?
- **Technical complexity:** Are required tools, data and infrastructure available?
- **Risk:** What technical, privacy, security or operational risks exist?
- **Scalability:** Can it support more users or larger workloads?
- **Accessibility:** Can different users use it effectively?
- **Impact:** How much improvement can it create?

**Impact–Effort prioritization:** Compare ideas using Impact and Effort. High-impact/low-effort ideas are often strong candidates for early implementation. High-impact/high-effort ideas may require planning. Low-impact ideas may be postponed or rejected.

## 11. Complete Unit 2 Flow

**Problem / User Need**
↓
**Empathize** – understand the user
↓
**Define** – clearly frame the problem
↓
**Divergent Thinking** – generate many possibilities
↓
**Mind Mapping / Brainstorming / Brainwriting / SCAMPER / Lateral Thinking**
↓
**Convergent Thinking** – evaluate alternatives
↓
**Select & Prioritize** – choose the most suitable ideas

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**Prototype / Implement / Test**

**Exam-ready conclusion:** Creative thinking helps designers move beyond conventional solutions. Divergent thinking expands possibilities, while convergent thinking narrows them using defined criteria. Mind mapping, brainstorming, brainwriting, SCAMPER and lateral thinking provide structured ways to generate ideas. In design thinking, Empathize and Define establish the user problem, Ideate produces possible solutions, and Selecting & Prioritizing identifies ideas that offer the best balance of user value, feasibility and constraints.

## 12. Important Short Questions

**1. Define creative thinking in computational design.**

**2. Differentiate between divergent and convergent thinking.**

**3. What is mind mapping?**

**4. State the main principles of brainstorming.**

**5. What is brainwriting?**

**6. Differentiate brainstorming and brainwriting.**

**7. What is SCAMPER?**

**8. Explain Substitute, Combine and Adapt in SCAMPER.**

**9. Explain Modify, Put to Another Use, Eliminate and Reverse/Rearrange.**

**10. What is lateral thinking?**

**11. Explain the Empathize stage.**

**12. What is a problem statement in the Define stage?**

**13. What is the purpose of the Ideate stage?**

**14. What is the difference between selecting and prioritizing ideas?**

**15. List criteria used for evaluating ideas.**

## 13. Key Terms for Revision

**Creative Thinking:** Generating novel and useful possibilities.
**Divergent Thinking:** Expanding possibilities by generating many ideas.
**Convergent Thinking:** Narrowing possibilities by evaluating and selecting ideas.
**Mind Mapping:** Visual organization of connected ideas around a central concept.
**Brainstorming:** Group verbal idea generation with deferred judgement.
**Brainwriting:** Written idea generation by participants.
**SCAMPER:** Substitute, Combine, Adapt, Modify/Magnify/Minify, Put to Another Use, Eliminate, Reverse/Rearrange.
**Lateral Thinking:** Exploring non-standard perspectives and patterns.
**Empathize:** Understand users.
**Define:** Frame the problem clearly.
**Ideate:** Generate possible solutions.
**Select:** Evaluate and choose suitable ideas.
**Prioritize:** Rank ideas according to value, feasibility and constraints.